

Kyle Main

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Senior Machine Learning Engineer & Security Data Scientist — Anomaly Detection, Clustering & Detection-as-Code

Security data scientist and ML engineer with 12 years of experience building unsupervised and time-series anomaly detection systems that run in production against massive-scale security telemetry — clustering network devices and entities by behavior, detecting anomalous authentication and process-chain patterns, and running exploratory analysis at scale on PySpark/Dataproc/BigQuery. Built and operate a multi-SIEM detection-as-code CI/CD pipeline (GitLab) with automated testing, staged/safe rollout, and rule-quality metrics tracking before production — the same engineering rigor this role asks for, already applied at scale (2,300+ rules across the MITRE ATT&CK matrix). M.S. Physics underpins the quantitative foundation.

CORE SKILLS

Machine Learning & Data Science: Clustering/unsupervised ML for device and entity behavior; time-series anomaly detection (authentication volume/velocity, process-parent/child chains); exploratory data analysis at scale; Python, PyTorch, scikit-learn, pandas, NumPy, SciPy

Big Data & Distributed Compute: PySpark, SparkSQL, GCP Dataproc, BigQuery, Apache Beam/Dataflow; working familiarity with Kafka and Flink streaming; cloud: GCP, AWS, Azure

MLOps / Detection-as-Code Pipelines: GitLab CI/CD pipeline for detection-as-code with automated unit/integration tests, staged/safe rollout, and formally tracked rule-quality metrics (coverage, precision/false-positive rate) before production deployment; multithreaded orchestration across customer environments

Containers & Orchestration: Docker (built and managed custom images; containerized reproducible detection-testing environments); hands-on experience operating within Kubernetes-orchestrated platforms

Cybersecurity Domain Depth: MITRE ATT&CK-mapped detection content (2,300+ rules); multi-SIEM API orchestration (Splunk, Microsoft Sentinel/Defender, Google SecOps, CrowdStrike, SentinelOne, Sumo Logic, XSIAM, Devo, ArcSight); threat intel integration into detection logic

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Design and manage statistical and behavioral detection analytics (Splunk saved searches) directly supporting Treasury's Security Operations Center incident response and case work (Treasury SOC / TSSOC, current project).
- Architected an unsupervised UEBA (User and Entity Behavior Analytics) detection layer from scratch for DOE/NNSA — Elasticsearch transforms baselining entity behavior across CrowdStrike, Suricata, and Zeek telemetry, plus anomaly-surfacing content, data-quality monitoring, and custom dashboards (DOE/NNSA Security Data Integration, completed) — turning a vague threat-visibility mandate into a concrete production system.
- Supported data ingestion and data-quality efforts within an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program (completed).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building statistical, behavioral, time-series, and ML-based (clustering) detection models against massive-scale customer telemetry, with a direct mandate to reduce alert fatigue and improve analyst trust.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built unsupervised clustering models to group devices on the network by behavioral feature for anomaly surfacing, and time-series anomaly detection covering authentication behavior (anomalous volume/attempt patterns by country over time) and process-chain behavior (parent/child process combinations), as core production detection content.
- Ran exploratory data analysis at scale on GCP Dataproc (Zeppelin notebooks, a home-grown reusable analysis toolkit, PySpark/SparkSQL) to mine 220+ ingested log sources for signal.
- Created and managed 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix as a very early hire, building the rules engine and detection content for the startup from scratch — plus the data engineering (50+ Logstash filters, Common Information Model) underneath it.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Developed custom statistical models and algorithms for DNS-based detection and mitigation of malware infections on the network, and analyzed large-scale security log data to surface anomalous behavior.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox